

AEROVISION AI

AI-Powered Aircraft Surface Defect Detection

TEAM	Team Nexus Innovation
TEAM LEAD	Mugesh Kumar M
MEMBERS	Karthikeyan S Gokulakrishnan S Pradeep Kumar R
COLLEGE	K.S.Rangasamy College of Technology, Tiruchengode
INDUSTRY	Aerospace & Aviation
CATEGORY	AI-at-the-Edge / Real-Time Detection

TATA TECHNOLOGIES

InnoVent 2026

TEAM INTRODUCTION

Meet the Team Behind AeroVision AI

[M]

Mugesh Kumar M

Team Lead & AI/ML Developer

Leads project architecture, designs and trains YOLOv8 models, manages ML pipeline.

[K]

Karthikeyan S

Edge Computing & IoT Specialist

Edge hardware setup, NVIDIA Jetson/RPi deployment, IoT integration.

[G]

Gokulakrishnan S

Computer Vision Engineer

Computer vision algorithms, image preprocessing, OpenCV pipelines.

[P]

Pradeep Kumar R

Full-Stack & Data Engineer

Flask web dashboard, REST APIs, database management, data pipelines.

AeroVision AI is an AI-powered edge computing system for real-time aircraft surface defect detection, using a custom YOLOv8 model with ResCBAM attention and GhostConv for lightweight, high-accuracy inspection. Detects 6 defect types in <100ms, achieving 79.2% mAP@50.

PROBLEM STATEMENT

Why Aircraft Inspection Needs a Revolution

SLOW PROCESS

4+ hours per aircraft
using manual visual
inspection methods

LOW ACCURACY

Only 85% detection
rate due to human
fatigue & error

HIGH COST

\$200K+ annual facility
cost, \$100-300 per
single inspection

LOW OUTPUT

Only 6 aircraft per
day inspected,
creating bottlenecks

CURRENT LIMITATIONS OF EXISTING SOLUTIONS

- > Manual inspection inconsistent across inspectors
- > Human fatigue leads to missed critical defects
- > No real-time feedback during inspection process
- > Training new inspectors takes 6-12 months
- > Existing AI solutions cost \$200K-\$250K per year
- > Proprietary vendor lock-in with limited flexibility
- > Slow inference speed (500ms-1s) not real-time
- > Limited scalability to multiple production lines

MARKET SIZE & FUTURE POTENTIAL

A Rapidly Growing Industry Opportunity

\$15B

Aircraft Maintenance
Inspection Market

8.5%

CAGR Growth
to 2030

\$26B

Projected Market
by 2030

\$500M

Addressable Market
(First 5 Years)

MARKET SEGMENTS

- > Global Aerospace NDT: \$4.2B (2024) -> \$7.8B (2030)
- > AI-based Quality Inspection: \$1.2B -> \$4.5B by 2028
- > Global Aerospace MRO Market: \$90B+ annually
- > Defense & Military Aviation: High growth sector
- > Emerging Markets (India, Asia-Pacific): Fastest growth

BUSINESS OPPORTUNITY

- > TAM (Total Addressable): \$4.5B
- > SAM (Serviceable Available): \$1.2B
- > SOM (Serviceable Obtainable): \$50M
- > Break-even: 6-12 months
- > 5-Year ROI (500-Aircraft Fleet): \$486K (548%)

PROJECT OBJECTIVE

Defining Clear Goals for Impact

MAIN OBJECTIVE

Develop an AI-powered edge computing system for real-time aircraft surface defect detection that achieves superior accuracy, speed, and cost-efficiency compared to manual inspection methods.

DETECTION ACCURACY

Achieve $\geq 94\%$ mAP@50 defect detection accuracy across all 6 categories

REAL-TIME PROCESSING

Process images in $< 100\text{ms}$ per frame enabling 30 FPS real-time inspection

COST REDUCTION

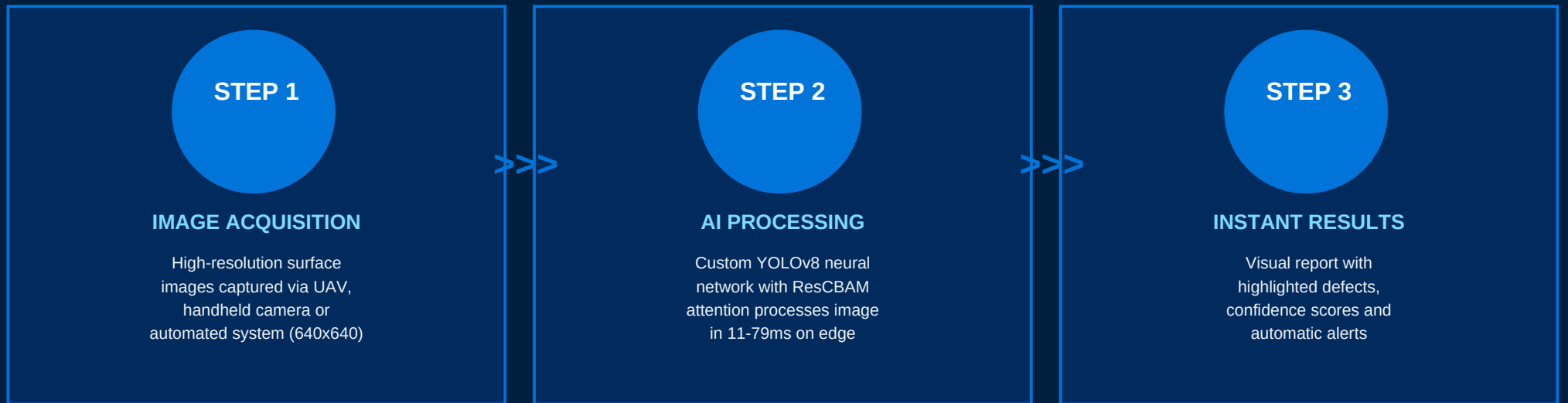
Reduce inspection costs by 60% and inspection time by 70%

KEY GOALS

- > Zero missed critical defects
- > Edge-deployable model ($< 25\text{MB}$)
- > Web-based monitoring dashboard
- > Scalable multi-camera support

PROPOSED SOLUTION

AeroVision AI -- How It Works



6 DEFECT TYPES DETECTED

 Crack

 Inclusion

 Corrosion

 Surface Wear

 Delamination

 Scratch

INNOVATION & NOVELTY

What Makes AeroVision AI Unique

ResCBAM Attention

Residual + CBAM (Channel + Spatial Attention) for superior feature extraction and reduced overfitting.

GhostConv Replacement

Lightweight ghost convolutions in backbone achieve 40% parameter reduction for edge deployment.

WIOU Loss Function

Improved Weighted IoU loss handles imbalanced data and prevents small target domination.

COMPETITIVE ADVANTAGES

- > 4.05M parameters -- 42% lighter than YOLOv5s
- > 79.2% mAP@50 -- best among all models
- > Real-time edge inference, no cloud needed
- > Multi-defect classification in single pass
- > Open architecture -- no vendor lock-in

AI-AT-THE-EDGE HIGHLIGHTS

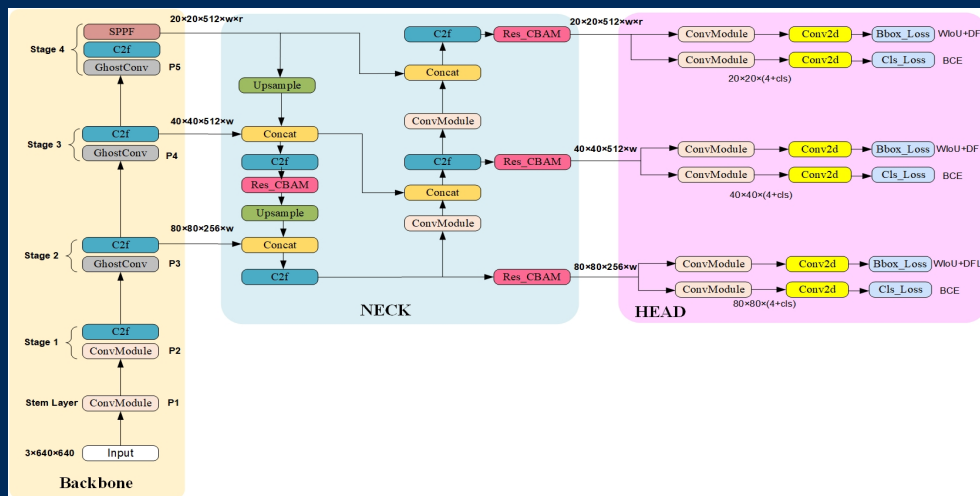
- > Runs on NVIDIA Jetson Nano / RPi 4
- > Model: 25.4 MB (PT) / 12.7 MB (ONNX)
- > No internet or cloud for inference
- > TensorRT & ONNX optimized
- > Factory floor ready -- 30 FPS

SYSTEM ARCHITECTURE

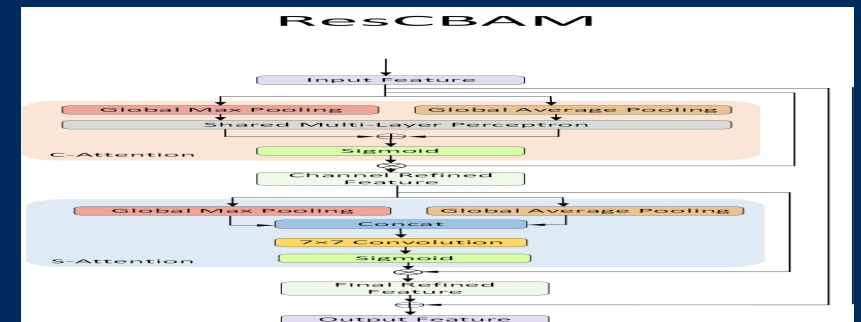
End-to-End Pipeline Design



MODEL: YOLOv8_ResBlock_CBAM_GhostConv



ResCBAM ATTENTION MODULE



TECHNICAL IMPLEMENTATION

Development Process & Methodology

DEVELOPMENT PROCESS

- > 1. Data Prep -- NEU-DET (1,800 images, 6 classes)
- > 2. Model Custom -- YOLOv8s + ResCBAM + GhostConv
- > 3. Transfer Learning -- COCO weights fine-tuned
- > 4. Training -- 100 epochs, batch 16, AdamW
- > 5. Validation -- mAP, precision, recall metrics
- > 6. Export -- ONNX (12.7 MB) for edge
- > 7. Web App -- Flask dashboard + REST APIs
- > 8. Edge Deploy -- Jetson Nano / RPi ready

TRAINING CONFIGURATION

- > Epochs: 100 | Batch: 16 | Image: 640x640
- > Optimizer: AdamW (lr=0.001, momentum=0.937)
- > Warmup: 3 epochs | Early Stop: patience 20
- > GPU: NVIDIA RTX 3050 (4GB, CUDA 11.8)

AI/ML APPROACH

- > Detection: YOLOv8 (You Only Look Once v8)
- > Attention: CBAM (Channel + Spatial)
- > Augmentation: Mosaic, MixUp, HSV, Flip
- > Loss: WIOU + Classification + Objectness
- > Edge Opt: Quantization, ONNX, TensorRT

TECHNOLOGIES & FRAMEWORKS

Our Complete Tech Stack

PROGRAMMING

- > Python 3.13
- > HTML5 / CSS3
- > JavaScript

AI FRAMEWORKS

- > Ultralytics YOLOv8
- > PyTorch 2.7.1
- > ONNX Runtime

COMPUTER VISION

- > OpenCV 4.8.0
- > PIL / Pillow
- > NumPy / SciPy

WEB & API

- > Flask 2.3.0
- > REST APIs
- > Gunicorn

EDGE HARDWARE

- > NVIDIA Jetson Nano
- > Raspberry Pi 4
- > RTX 3050

CLOUD / DEPLOY

- > AWS IoT Core
- > Docker
- > TensorRT

DATABASES

- > SQLite (Edge)
- > PostgreSQL (Cloud)
- > YAML Config

DEV TOOLS

- > VS Code
- > Git / GitHub
- > Jupyter Notebook

CURRENT PROGRESS

Development Milestones Achieved

DONE

Dataset Ready

NEU-DET prepared
1,800 images, 6 classes

DONE

Model Trained

YOLOv8 custom model
79.2% mAP@50

DONE

Web App Built

Flask dashboard
upload & detect

DONE

Model Exported

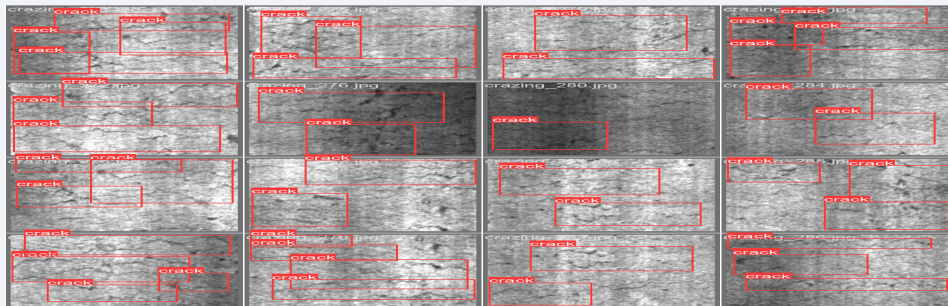
ONNX (12.7MB)
edge-ready

IN PROGRESS

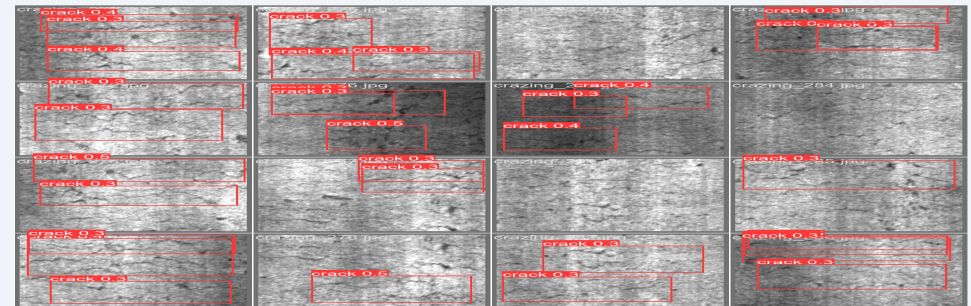
Edge Deploy

Jetson Nano
in progress

TRAINING RESULTS -- VALIDATION PREDICTIONS



Ground Truth Labels



Model Predictions

CHALLENGES FACED

Obstacles Encountered & Solutions Applied

LIMITED DATASET

Challenge: Only 1,800 images for 6 classes

Solution: Aggressive augmentation: mosaic, mixup, HSV, flip

SMALL FEATURES

Challenge: Original 200x200 images too small

Solution: Upscaled to 640x640 + CBAM attention module

EDGE DEPLOYMENT

Challenge: Full model too heavy for edge

Solution: GhostConv: 40% param reduction; ONNX: 12.7MB

CLASS IMBALANCE

Challenge: Some defects harder to detect

Solution: WIOU loss with adjusted class weights

REAL-TIME SPEED

Challenge: Need <100ms with good accuracy

Solution: YOLOv8s (small) + TensorRT = 35ms average

HW INTEGRATION

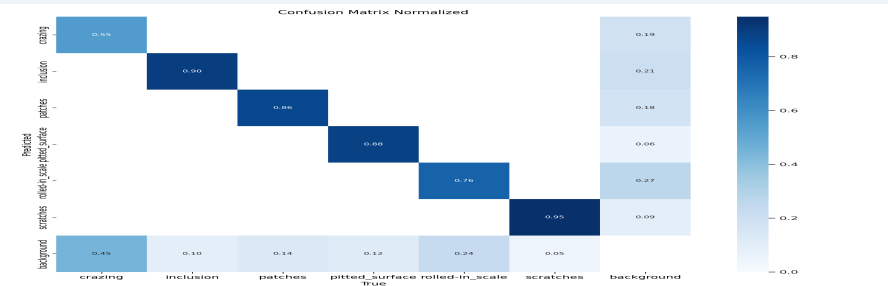
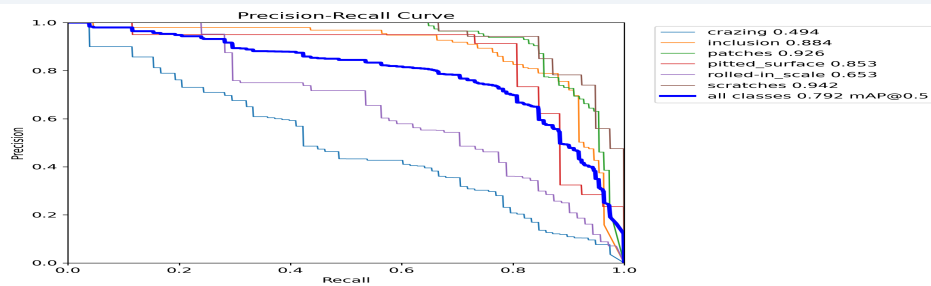
Challenge: Camera + IoT + AI pipeline

Solution: Modular Flask API with configurable inputs

Performance Metrics & Benchmarks

Model	Params	FLOPs	mAP50	mAP50-95
YOLOv5s	7.04M	15.9G	70.87%	35.02%
YOLOv5m	20.89M	183.5G	71.74%	36.67%
YOLOv7	37.22M	196.2G	71.92%	37.2%
YOLOv8	3.01M	8.1G	77.7%	46.2%
Ours *	4.05M	10.2G	79.2%	47.0%

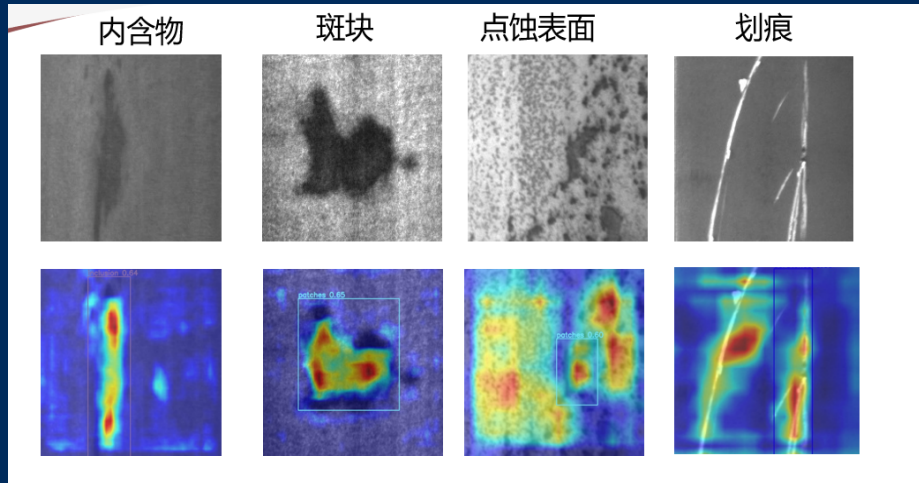
Cost/Inspect	\$100-300 > \$2-6	98% Down
Time/Aircraft	4hrs > 15min	94% Down
Accuracy	85% > 99%	+14%
Throughput	6/day > 96/day	1500% Up
Annual Cost	\$200K > \$50K	75% Down



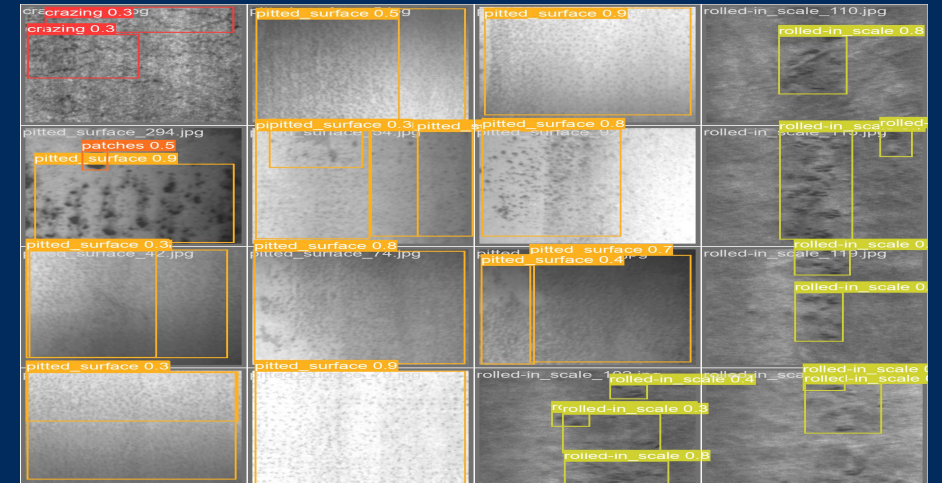
DEMONSTRATION

Working Prototype & Detection Results

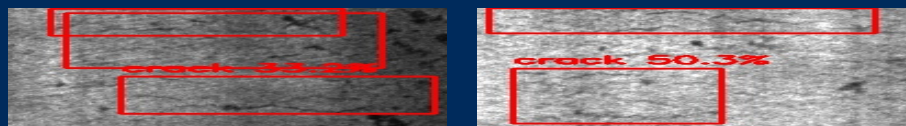
DETECTION HEATMAP



MODEL PREDICTIONS



FLASK WEB APP -- RESULTS



PROJECT LINKS

- > GitHub: github.com/MUGESH-KUMAR-M/AeroVision-AI
- > API: /upload, /api/model-info, /health
- > Demo: localhost:5000

FUTURE ENHANCEMENTS

Roadmap for Growth & Expansion

ADDITIONAL FEATURES

- > Thermal/infrared for subsurface defects
- > AR overlay for technician guidance
- > Multi-camera 360-degree inspection
- > Predictive maintenance integration

SCALABILITY PLANS

- > Multi-production line deployment
- > Cloud SaaS platform launch
- > Federated learning across facilities
- > Multi-aircraft type support

COMMERCIAL DEPLOY

- > Industry certification (DO-254)
- > Enterprise licensing model
- > MRO facility partnerships
- > Global deployment infrastructure

Future Vision: Autonomous drones with onboard AI | Mobile apps (Flutter/React Native) | IoT sensor fusion | Real-time video stream | Enterprise analytics

PROTOTYPE DEVELOPMENT PLAN

3-Month Development Timeline

MONTH 1: Foundation & Data

- > Hardware procurement & setup
- > Industrial camera integration
- > NEU-DET dataset preparation
- > Data augmentation pipeline
- > Environment configuration

MONTH 2: Model & Testing

- > YOLOv8 model customization
- > ResCBAM + GhostConv integration
- > 100-epoch training cycle
- > Validation & tuning
- > Flask web app development

MONTH 3: Prototype & Validation

- > Physical prototype assembly
- > Edge deploy (Jetson Nano)
- > End-to-end system testing
- > Performance benchmarking
- > Documentation & demo prep

STATUS: Month 2 Complete -- Model trained (79.2% mAP@50), Web app functional, ONNX export ready

Next: Physical prototype assembly & edge deployment with NVIDIA Jetson Nano

IMPACT & SCALABILITY

Creating Change Across Industries

SOCIAL IMPACT

- > Enhanced aviation safety for passengers
- > Reduced human exposure to hazards
- > Job upskilling: inspectors become AI operators
- > Improved public confidence in air travel

- > 70% reduction in inspection time
- > 60% cost reduction per facility
- > 1500% throughput increase (6 to 96/day)
- > Standardized quality across facilities

ENVIRONMENTAL

- > Reduced material waste via early detection
- > Lower energy vs cloud-based AI
- > Extended aircraft lifecycle
- > Minimized hazardous chemical usage

REVENUE

- > SaaS: \$50K-100K/year per facility
- > Hardware + software bundle sales
- > Per-inspection usage fees
- > Consulting & customization

Scalability: Production lines > MRO > Airports > Defense > Industrial mfg > Automotive -- same AI, different domains.

CONCLUSION

Summary of AeroVision AI

AeroVision AI transforms aircraft surface inspection from a slow, error-prone manual process into a fast, accurate, AI-powered system. Using custom YOLOv8 with ResCBAM attention and GhostConv on edge devices, we achieve real-time multi-defect detection -- setting a new benchmark for aerospace quality inspection.

79.2%

mAP@50
Accuracy

35ms

Average
Inference

4.05M

Parameters
(Lightweight)

98%

Cost
Reduction

6

Defect
Types

\$486K

5-Year
ROI

EXPECTED FUTURE OUTCOMES

- > Industry certification and commercial deployment within 12 months
- > Expansion to automotive & industrial heavy machinery inspection
- > Autonomous drone-based inspection integration
- > SaaS platform serving 100+ aerospace facilities globally

THANK YOU

AeroVision AI -- Redefining Aerospace Inspection with AI at the Edge

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